

Taller práctico: Operadores y expresiones Programación

$$\begin{aligned}
 a) & 25 - (10 - 2^3) * 4 / 2 + 15 \% 4 \\
 & 25 - (10 - 8) * 4 / 2 + 15 \% 4 \\
 & 25 - 2 * 4 / 2 + 15 \% 4 \\
 & 25 - 8 / 2 + 15 \% 4 \\
 & 25 - 4 + 3 \\
 & 21 + 3 \\
 & = 24
 \end{aligned}$$

$$\begin{array}{r}
 15 \overline{) 4} \\
 3 \overline{) 3} \\
 \hline
 5
 \end{array}$$

$$\begin{aligned}
 b) & (8+2) * (5-3)^2 / (10 \% 3 + 1) \\
 & 10 * 2^2 / (1 + 1) \\
 & 10 * 2^2 / 2 \\
 & 10 * 4 / 2 \\
 & 40 / 2 \\
 & = 20
 \end{aligned}$$

$$\begin{array}{r}
 10 \overline{) 3} \\
 1 \overline{) 3} \\
 \hline
 3
 \end{array}$$

$$\begin{aligned}
 c) & \text{trunc}(45/2) + 5 * (-10+6)^2 \\
 & \text{trunc}(45/2) + 5 * -4^2 \\
 & \text{trunc}(45/2) + 5 * 16 \\
 & 22 + 80 \\
 & = 102
 \end{aligned}$$

$$\begin{array}{r}
 45 \overline{) 2} \\
 05 \overline{) 2} \\
 \hline
 1
 \end{array}$$

$$d) 50 - 3 * (5 \% 2) + (18 / 3) ^ 2 - 4$$

$$50 - 3 * 1 + 6 ^ 2 - 4$$

$$50 - 3 * 1 + 36 - 4$$

$$50 - 3 + 36 - 4$$

$$47 + 32$$

$$= 79$$

$$e) (20 \% 6) * 3 + \text{trunc}(15 / 4) * 2 ^ 3$$

$$2 * 3 + \text{trunc}(15 / 4) * 8$$

$$2 * 3 + \text{trunc}(15 / 4) * 8$$

$$6 + 3 * 8$$

$$6 + 24$$

$$= 30$$

Parte 2. Operadores relacionales y lógicos combinados

$x=12$ $y=5$ $z=8$ $w=15$

$$a) (x \% y == 2) \text{ AND } (z * 2 >= w) \text{ OR } ! (x > w)$$

$$(12 \% 5 == 2) \text{ AND } (8 * 2 >= 15) \text{ OR } ! (12 > 15)$$

$$(2 == 2) \text{ AND } (16 >= 15) \text{ OR } ! (12 > 15)$$

$$(V) \text{ AND } (V) \text{ OR } ! (F)$$

$$(V) \text{ OR } ! (F)$$

$$V \text{ OR } V$$

Verdadero

$$b) ! ((x / 3 > 2) \text{ AND } (w \% z == 7)) \text{ OR } (y ^ 2 <= x * 2)$$

$$! ((12 / 3 > 2) \text{ AND } (15 \% 8 == 7)) \text{ OR } (5 ^ 2 <= 12 * 2)$$

$$! ((4 > 2) \text{ AND } (7 == 7)) \text{ OR } (25 <= 24)$$

$$! ((V) \text{ AND } (V)) \text{ OR } (F)$$

$$! (V) \text{ OR } (F)$$

$$(F) \text{ OR } (F)$$

False

$$c) (w - x + y == 8) \text{ AND } ((z * 4) / 10 == 4)$$

$$(15 - 12 + 5 == 8) \text{ AND } ((8 * 5) / 10 == 4)$$

$$(15 - 12 + 5 == 8) \text{ AND } (40 / 10 == 4)$$

$$(15 - 12 + 5 == 8) \text{ AND } (4 == 4)$$

$$(8 == 8) \text{ AND } (4 == 4)$$

$$(V) \text{ AND } (V)$$

$$= \text{verdadeiro}$$

$$d) (x \% 2 == 0) \text{ AND } !(y \% 2 == 0) \text{ AND } (w \% 4 == 0)$$

$$(12 \% 2 == 0) \text{ AND } !(5 \% 2 == 0) \text{ AND } (15 \% 4 == 0)$$

$$(0 == 0) \text{ AND } !(1 == 0) \text{ AND } (0 == 0)$$

$$(V) \text{ AND } (F) \text{ AND } (V)$$

$$(V) \text{ AND } (V) \text{ AND } (V)$$

$$(V) \text{ AND } (V)$$

$$= \text{verdadeiro}$$

$$e) (x + y > w) \text{ AND } (z^2 < x * w)$$

$$(12 + 5 > 15) \text{ AND } (8^2 < 12 * 15)$$

$$(12 + 5 > 15) \text{ AND } (64 < 12 * 15)$$

$$(17 > 15) \text{ AND } (64 < 180)$$

$$(V) \text{ AND } (V)$$

$$= \text{verdadeiro}$$

$$f) !(w \% 4 == 0) \text{ OR } (x / 2 != 6) \text{ AND } (z + y == 13)$$

$$!(15 \% 4 == 0) \text{ OR } (12 / 2 != 6) \text{ AND } (8 + 5 == 13)$$

$$!(0 == 0) \text{ OR } (6 != 6) \text{ AND } (13 == 13)$$

$$(F) \text{ OR } (F) \text{ AND } (V)$$

$$(F) \text{ OR } (F) \text{ AND } (V)$$

$$\text{Falso AND } (V)$$

$$\text{Falso}$$

$$g) ((w - 2) * 2 >= x) \text{ AND } !(y^2 == 25)$$

$$((15 - 2) * 2 >= 12) \text{ AND } !(5^2 == 25)$$

$$(13 * 2 >= 12) \text{ AND } !(25 == 25)$$

$$(26 >= 12) \text{ AND } !(25 == 25)$$

$$(V) \text{ AND } (F)$$

$$= \text{Falso}$$

h) $(x * y == 60) \text{ OR } !((w / y == 3) \text{ AND } (z \% 3 == 2))$
 $(12 * 5 == 60) \text{ OR } !((15 / 5 == 3) \text{ AND } (8 \% 3 == 2))$
 $(60 == 60) \text{ OR } !((3 == 3) \text{ AND } (2 == 2))$
 $(V) \text{ OR } !((V) \text{ AND } (V))$
 $(V) \text{ OR } !(V)$
 verdadero

i) $!(x < z) \text{ AND } ((w + y) / 4 == 4) \text{ OR } (z * 2 == 16)$
 $!(12 < 8) \text{ AND } ((15 + 5) / 4 == 5) \text{ OR } (8 * 2 == 16)$
 $!(12 < 8) \text{ AND } (20 / 4 == 5) \text{ OR } (16 == 16)$
 $!(12 < 8) \text{ AND } (5 == 5) \text{ OR } (16 == 16)$
 $!(F) \text{ AND } (V) \text{ OR } (V)$
 $(V) \text{ AND } (V) \text{ OR } (V)$
 verdadero OR (V)
 verdadero

j) $((x \% 3 == 0) \text{ OR } (y \% 2 == 1)) \text{ AND } !(w - 2 == 7)$
 $((12 \% 3 == 0) \text{ OR } (5 \% 2 == 1)) \text{ AND } !(15 - 2 == 7)$
 $((0 == 0) \text{ OR } (1 == 1)) \text{ AND } !(13 == 7)$
 $(V) \text{ OR } (V) \text{ AND } !(V)$
 $(V) \text{ AND } (F)$
 Falso

Parte 3. Trabajo de algoritmo (Prueba de Escritorio)

Ejercicio A

	instrucción	A	B	C
1.	$A \leftarrow 10$	10	-	-
2.	$B \leftarrow 5$	10	5	-
3.	$C \leftarrow A * 2$	10	5	100
4.	$A \leftarrow C / B$	20	5	100
5.	$B \leftarrow B \% 2 + A$	20	21	100
6.	$C \leftarrow C - A * B$	20	21	-320
7.	$A += 5$	25	21	-320
8.	$B --$	25	20	-320

Ejercicio B

1. $\text{grupos} \leftarrow 25$
2. $\text{horaDirecta} \leftarrow 40$
3. $\text{totalHora} \leftarrow \text{grupos} * \text{horaDirecta}$
4. $\text{grupos} += 5$
5. $\text{horaDirecta} -= 10$
6. $\text{totalHora} \leftarrow \text{totalHora} + (\text{grupos} * \text{horaDirecta})$
7. $\text{grupos} ++$
8. $\text{totalHora} \% = 3$

Instrucción	grupos	horaDirecta	totalHora	
1. $\text{grupos} \leftarrow 25$	25	-	-	25
2. $\text{horaDirecta} \leftarrow 40$	25	40	-	40
3. $\text{totalHora} \leftarrow \text{grupos} * \text{horaDirecta}$	25	40	1000	1000
4. $\text{grupos} += 5$	30	40	1000	
5. $\text{horaDirecta} -= 10$	30	30	1000	30
6. $\text{totalHora} \leftarrow \text{totalHora} + (\text{grupos} * \text{horaDirecta})$	30	30	1900	900 + 1000
7. $\text{grupos} ++$	31	30	1900	
8. $\text{totalHora} \% = 3$	31	30	1	1900 3 10 633 10

Ejercicio C

Instrucción	registros	capacidadLote	lotesCompleto	Sobrante
1. $\text{registros} \leftarrow 150$	150	-	-	-
2. $\text{capacidadLote} \leftarrow 40$	150	40	-	-
3. $\text{lotesCompleto} \leftarrow \text{trunc}(\text{registros} / \text{capacidadLote})$	150	40	3	-
4. $\text{sobrante} \leftarrow \text{registros} \% \text{capacidadLote}$	150	40	3	30
5. $\text{registros} += 15$	165	40	3	30
6. $\text{lotesCompleto} \leftarrow \text{trunc}(\text{registros} / \text{capacidadLote})$	165	40	4	30
7. $\text{sobrante} \leftarrow \text{registros} \% \text{capacidadLote}$	165	40	4	5
8. $\text{capacidadLote} -= 1$	165	39	4	5

③ $150 \div 40$
30 3

④ $165 \div 40$

Ejercicio D

Instrucciones	Instructores	fichas	cargaBase	fichasSinAsignar
1. Instructores $\leftarrow 8$	8	-	-	-
2. fichas $\leftarrow 25$	8	25	-	-
3. cargaBase $\leftarrow \text{trunc}(\text{fichas} / \text{Instructores})$	8	25	3	-
4. fichasSinAsignar $\leftarrow \text{fichas} - \text{Instructores} \cdot \text{cargaBase}$	8	25	3	1
5. Instructores --	7	25	3	1
6. cargaBase $\leftarrow \text{trunc}(\text{fichas} / \text{Instructores})$	7	25	3	1
7. fichas $\leftarrow 2 \cdot 2$	29	25	3	1
8. fichasSinAsignar $\leftarrow \text{fichas} - \text{Instructores} \cdot \text{cargaBase}$	29	25	3	1